Tin Vu

Distributed Database Systems | Geospatial Computing | Query Optimization

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Seattle, Washington | No visa sponsorship required

PROFILE

Applied Scientist at Microsoft with a PhD in Computer Science from UC Riverside. Work focuses on distributed spatial databases, query optimization, spatial joins, and indexing, including learned approaches to partitioning, join optimization, and sampling.

EDUCATION

PhD, Computer Science - University of California, Riverside; advisor: Ahmed Eldawy 2021

BS, Computer Science - Hanoi University of Science and Technology 2013

EXPERIENCE

Microsoft | Applied Scientist 2021 - Present
Software Engineer Intern 2019 and 2020
Developed query understanding models to improve Bing Maps geocoding quality.

Environmental Systems Research Institute (Esri) | Software Development Engineer Intern 2019
In the ArcGIS GeoDatabase group, improved Utility Network performance and scalability using parallel processing. Earned 2nd Place overall and the Best Presentation Award at the Esri Intern Hackathon.

VNG Corporation | Data Engineer 2014 - 2016
Developed a database system for an online game service serving five million customers.

SELECTED PROJECTS

Optimized Spatial Joins in Apache AsterixDB. Implemented partition-based spatial merge joins, SQL++ spatial predicates, optimizer rules and partitioning hints, static and dynamic partitioning, and optimizer-plan and runtime tests. A local synthetic benchmark reduced median join time from 151.056 to 0.380 seconds against the same build with the spatial rewrite disabled.

R-Grove Spatial Indexes for Beast. Developed R-Grove, R*-Grove, and RR*-Grove indexes for Beast, a geospatial processing system built on Apache Spark. R*-Grove adapts R*-tree splitting and a balance factor to control partition sizes, and combines sampling with a storage-size histogram.

Incremental Spatial Partitioning. Developed a framework for bulk insertions and deletions with budgeted partition reorganization. Formalized the NP-hard partition-selection problem, implemented R*-tree, LSM-tree, and cost-based strategies, and evaluated query throughput and partition quality.

Learned Query Optimizer for Spatial Join. Developed models for join cardinality and candidate-comparison estimation, algorithm and partitioning selection, and execution parameter tuning. Evaluated the models against baselines on synthetic and real spatial datasets and released code and datasets.

Learning-Based Spatial Partitioning. Developed a model to select partitioning techniques from dataset summaries without executing each candidate method. Evaluated six techniques against four quality metrics using spatial histograms and measures of skew.

SELECTED PUBLICATIONS

- Geospatial Support in Apache AsterixDB - ACM SIGSPATIAL, 2026; **Best Industry Paper Award.**
- A Learning-Based Framework for Spatial Join Processing - The VLDB Journal, 2024.
- Incremental Partitioning for Efficient Spatial Data Analytics - PVLDB, 2022.
- R*-Grove: Balanced Spatial Partitioning for Large-Scale Datasets - Frontiers in Big Data, 2020.
- R-Grove: Growing a Family of R-Trees in the Big-Data Forest - ACM SIGSPATIAL, 2018.